

Education

University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies

Associate Software Engineer

Dallas, TX

June 2025 – Present

- Developed real-time signal processing algorithms in C++ and Python for defense sensor systems on embedded Linux hardware, applying coordinate transforms and DSP techniques to translate raw sensor streams into structured, actionable data.
- Built high-throughput data ingestion and processing microservices handling live sensor data under strict latency constraints, collaborating directly with electrical and firmware engineers to validate output against hardware behavior.
- Maintained automated CI/CD build infrastructure with static analysis and integration testing across multiple production deployments.

Doxy.me

Software Engineer

Charleston, SC

August 2024 – May 2025

- Built end-to-end ML data pipelines in Python using PyTorch and Llama to process, structure, and run inference over large volumes of unstructured clinical data captured via WebRTC video and audio streams.
- Deployed and operated inference infrastructure on AWS SageMaker and Fargate, iterating on model accuracy using reinforcement learning and owning the full pipeline from raw input to structured output.
- Authored YAML-based CI/CD configuration and cloud infrastructure automation, enabling reproducible environments and zero-downtime releases across multiple production services.

University of Utah

Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT

May 2024 – Jun 2025

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

Projects

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.